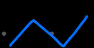
A2 Organic reactions

ISOMERS

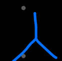
Structural

→ same molecular formula & different structural formula

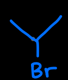
① chain


butane

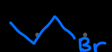
&


2-methylpropane

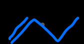
② position


2-bromopropane

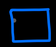
&


1-bromopropane

③ functional group


but-1-ene

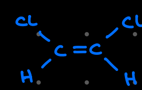
&

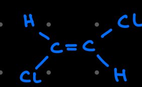

cyclic alkane

Stereoisomerism

→ same structural formula & different spatial arrangement

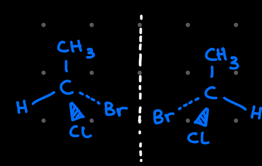
① geometrical (E-/Z-)


cis


trans

due to restricted rotation

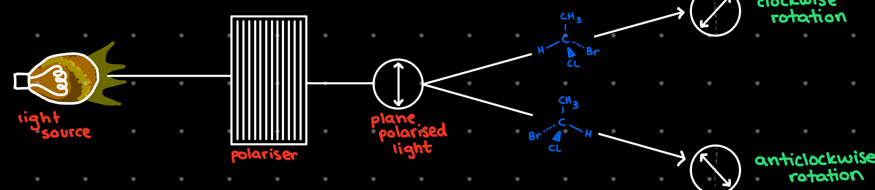
② optical



non-superimposable mirror images

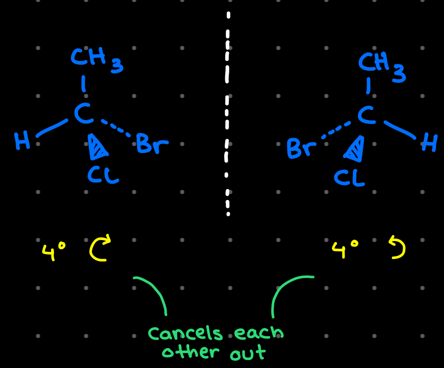
Optical isomerism:

↳ non-superimposable mirror images
↳ only in chiral compound (has a carbon attached to four different groups)
↳ can rotate plane polarised light



→ Racemic mixture

- a mixture containing equal amounts of both enantiomers of a chiral compound
- rotation of both enantiomers cancel each other out
- racemic mixture does not rotate plane polarised light
- it is optically inactive



Why does optical isomerism matter?

enantiomers

→ rotate plane-polarised light

→ can have different biological properties

1) different biological activity of enantiomers

enzymes/receptors are chiral so they can differentiate b/w enantiomers

thalidomide :

enantiomer 1

therapeutic effects

enantiomer 2

causes birth defects

→ separation of a racemic mixture

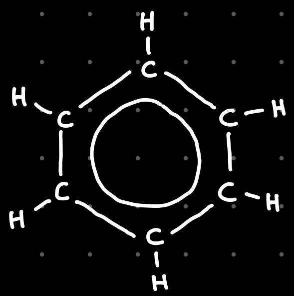
* enantiomers can have different effects; its necessary to separate a racemic mixture into its pure enantiomers

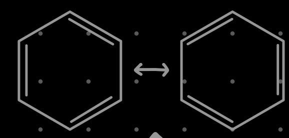
technique 1 : CHIRAL RESOLUTION separating enantiomers through their difference in interactions with other chiral substances

technique 2 : ENANTIOMER-SPECIFIC RESOLUTION developing synthetic routes that directly only produce desired enantiomers

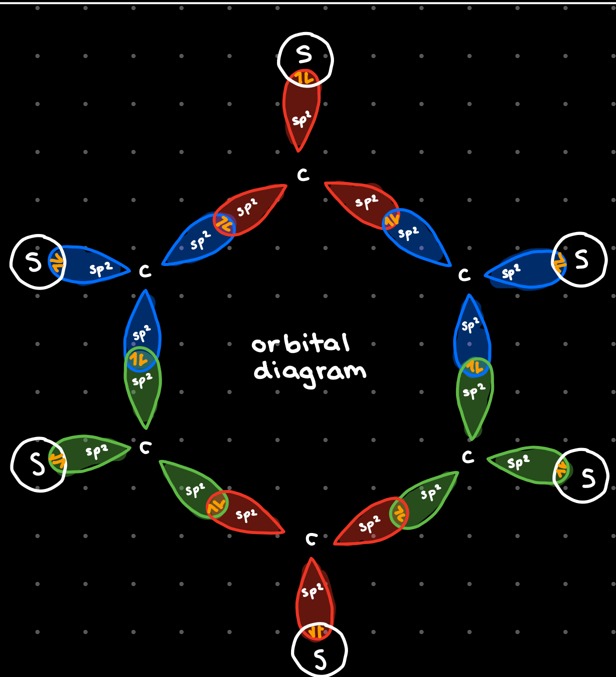
technique 3 : CHIRAL CATALYSTS are chiral themselves & can produce a single pure optical isomer

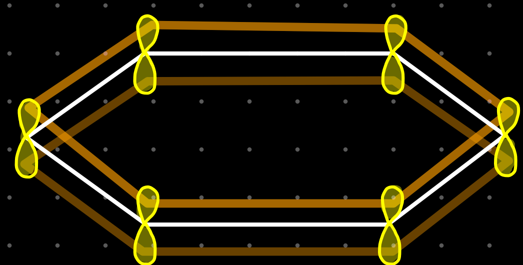
Structure of benzene (C_6H_6)





* regular hexagonal
* all C-C bond lengths are the same
* bond angle is 120°
* each carbon is sp^2 hybridised
* three alternate delocalised π bonds are present
* very stable molecule due to the delocalisation of π electrons
  ↳ thus undergoes substitution reaction easily but not addition reactions





each carbon atom in benzene ring has a unhybridised 2p orbital which forms π bond by sideways overlap

Reactions of benzene

1) Friedel-Craft alkylation
2) Friedel-Craft acylation
3) Halogenoation
4) Nitration
5) Halogenation

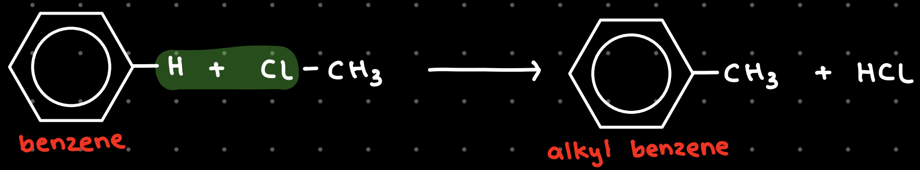
electrophilic substitution

1) Friedel-Craft alkylation

reagent : halogenoalkane

catalyst : $AlCl_3$ / $FeCl_3$

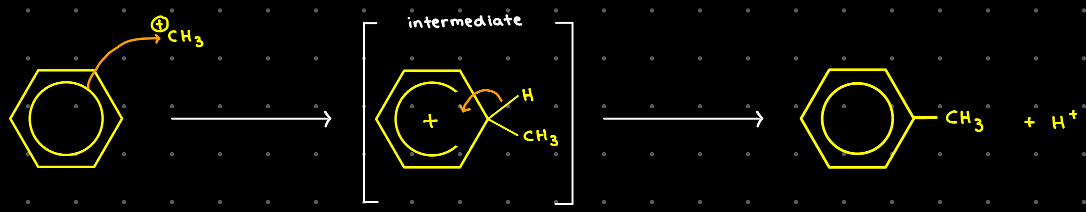
conditions : heat



reaction mechanism : ELECTROPHILIC SUBSTITUTION

generation of electrophile ⇒ $\overset{\delta+}{CH_3}-\overset{\delta-}{Cl}: + AlCl_3 \longrightarrow CH_3^+ + [AlCl_4]^-$

mechanism ⇒

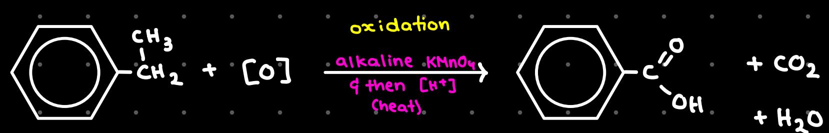


regeneration of catalyst ⇒ $AlCl_4^+ + H^+ \longrightarrow \underline{\underline{AlCl_3}} + HCl$

✱ formation of benzoic acid

① alkylation of benzene to make methylbenzene

② complete oxidation of side chain of methylbenzene

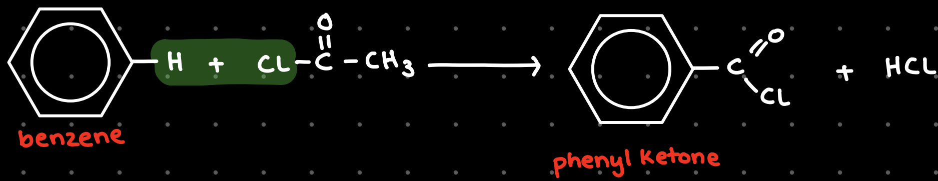


2) Friedel-Craft acylation

reagent : acyl chloride

catalyst : $AlCl_3 / FeCl_3$

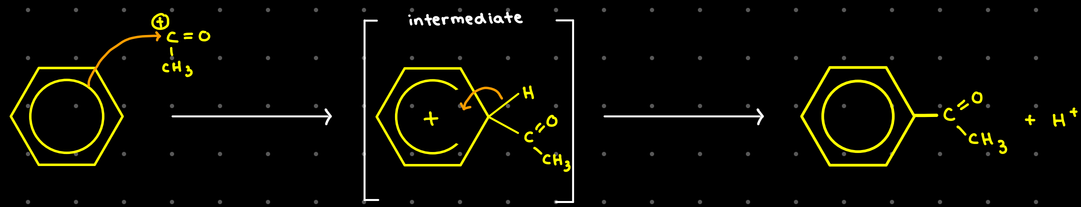
conditions : heat



reaction mechanism : ELECTROPHILIC SUBSTITUTION

generation of electrophile ⇒ $CH_3-\overset{\delta+}{C}(=O)-\overset{\delta-}{Cl}: + AlCl_3 \longrightarrow CH_3-C^{\oplus}(=O) + [AlCl_4]^-$

mechanism ⇒



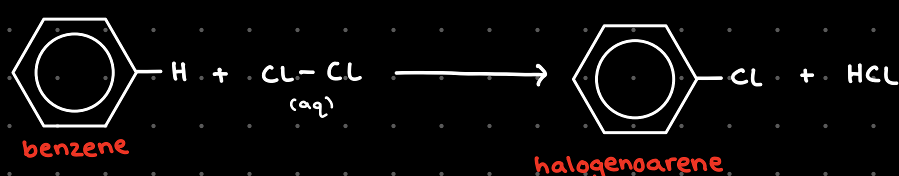
regeneration of catalyst ⇒ $AlCl_4^- + H^+ \longrightarrow \underline{\underline{AlCl_3}} + HCl$

3) Halogenation

reagent : Cl_2 , Br_2

catalyst : $AlCl_3 / AlBr_2$

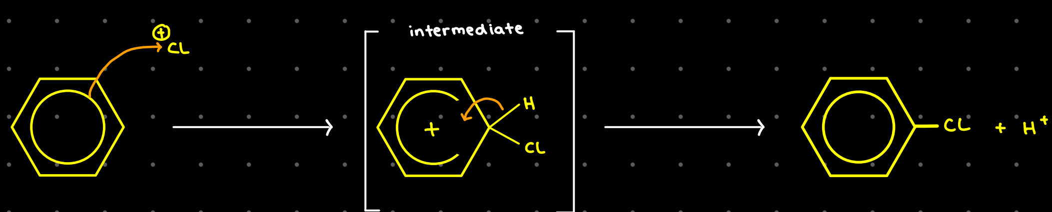
conditions : room temp. & anhydrous



reaction mechanism : ELECTROPHILIC SUBSTITUTION

generation of electrophile ⇒ $\overset{\delta+}{Cl}-\overset{\delta-}{Cl}: + AlCl_3 \longrightarrow Cl^+ + [AlCl_4]^-$

mechanism ⇒



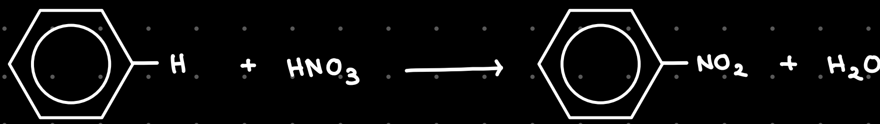
regeneration of catalyst ⇒ $AlCl_4^- + H^+ \longrightarrow \underline{\underline{AlCl_3}} + HCl$

4) Nitration

reagent : conc. HNO_3

catalyst : conc. H_2SO_4

conditions : 50° under reflex

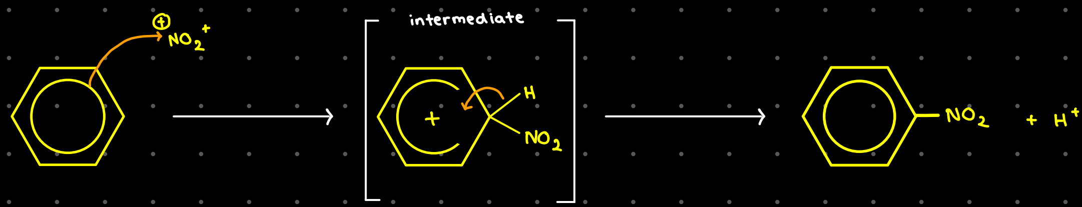


benzene nitrobenzene

reaction mechanism : ELECTROPHIC SUBSTITUTION

generation of electrophile ⇒ $H_2SO_4 + HNO_3 \longrightarrow HSO_4^- + NO_2^+ + H_2O$

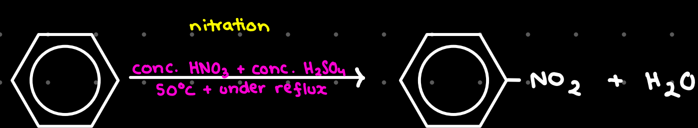
mechanism ⇒



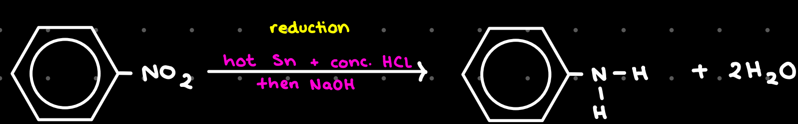
regeneration of catalyst ⇒ $HSO_4^- + H^+ \longrightarrow \underline{\underline{H_2SO_4}}$

*** formation of phenylamine**

① nitration of benzene to make nitrobenzene



② reduction of nitrobenzene to make phenylamine

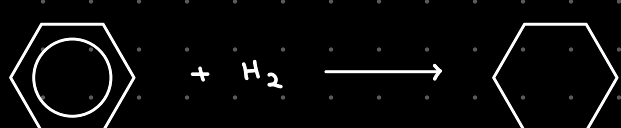


5) Hydrogenation

reagent : H_2

catalyst : Pt / Ni

conditions : heat & pressure



benzene cyclohexane

★ delocalised π system is broken & a saturated ring is formed

Phenol

↳ crystalline solid

↳ weakly acidic

∴ MORE acidic than ethanol

∴ Less acidic than metal carbonates

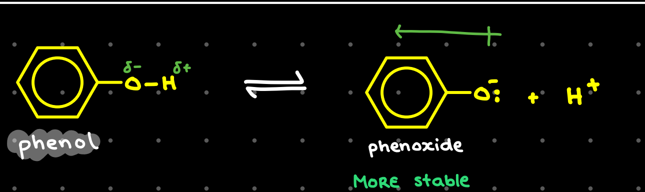
decreasing acidity

phenols > water > ethanol

WHY ?

decreasing stability

phenoxide > hydroxide > ethoxide



* oxygen atom in phenoxide ion has lone pair of e^-
* that lone pair of e^- becomes delocalised into ring
* charge is spread over the ion = MORE STABLE ion

$$CH_3-CH_2-\overset{\delta-}{\ddot{O}}-\overset{\delta+}{H} \rightleftharpoons CH_3-CH_2-\ddot{O}:^- + H^+$$

ethanol

MORE stable

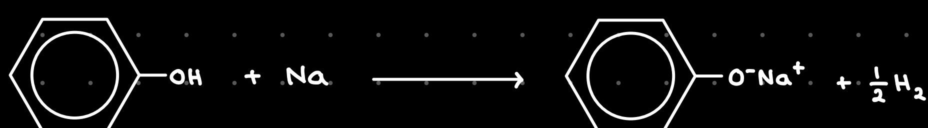
ethoxide

* alkyl group is electron-donating (strengthens O-H bond)
* ethanol is MORE STABLE than ethoxide

Reactions of phenols

1) reaction with Na
2) reaction with NaOH
3) nitration
4) bromination
5)

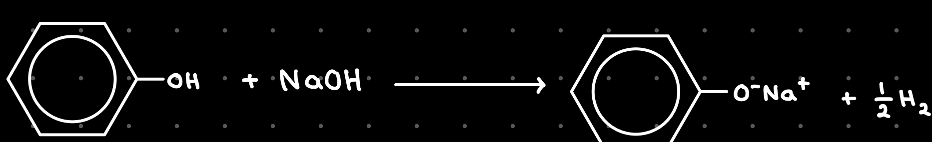
1) reaction with Na



phenol

sodium phenoxide

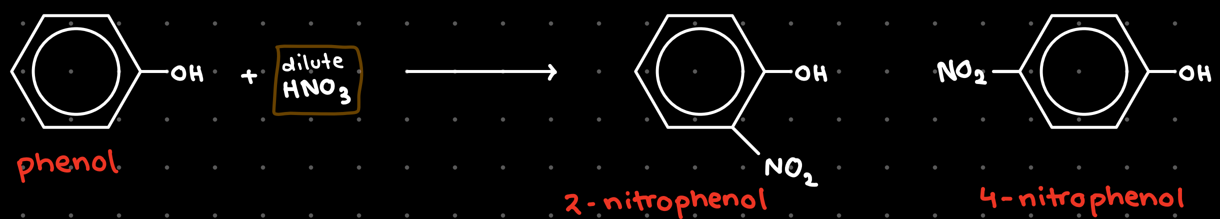
2) reaction with NaOH



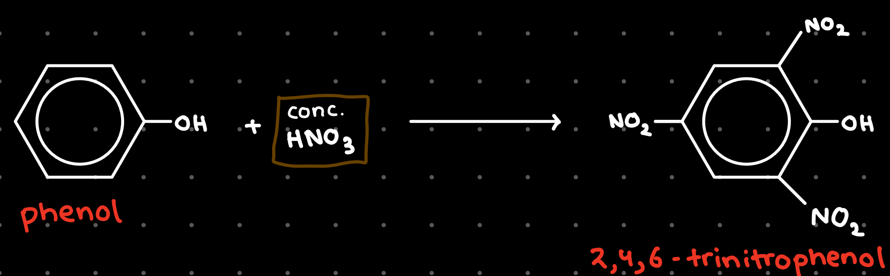
phenol

sodium phenoxide

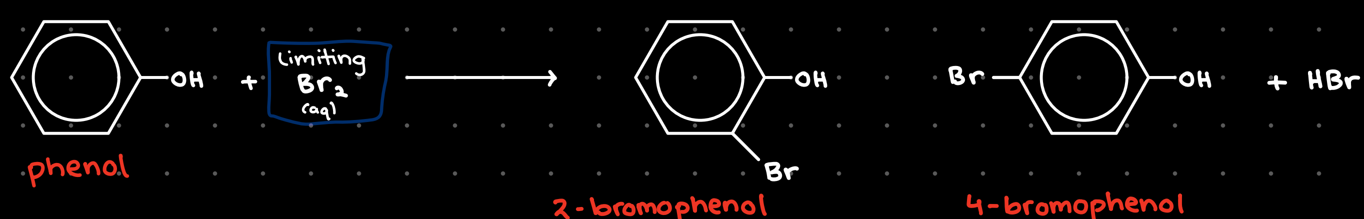
3) nitration



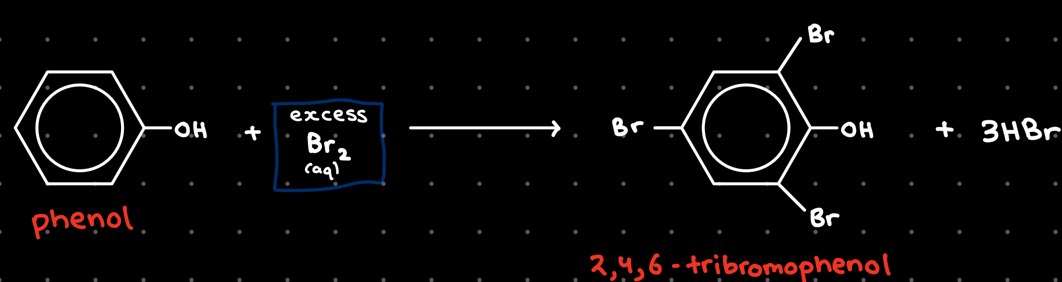
IF ~~conc.~~ HNO_3 is used →



3) bromination



IF ~~excess~~ Br_2 is used →

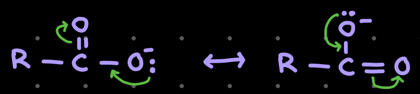


WHITE PPT.

Carboxylic acid & its derivatives

$R-\overset{O}{\overset{\|}{C}}-O-H$ — $C=O$ is electron-withdrawing so it weakens $O-H$ bond

once O-H bond is broken



carboxylate ion is stabilised by movement of lone pair in COO (resonance)

Oxidation of methanoic acid & ethanedioc acid

aldehyde group

$$\underset{\text{methanoic acid}}{H-\overset{O}{\overset{\|}{C}}-O-H} \xrightarrow[\text{acid. } KMnO_4 / \text{acid: } K_2Cr_2O_7]{\text{Fehling / Tollens' /}} CO_2 + H_2O$$

$$\underset{\text{ethanedioc acid}}{HO-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-OH} \xrightarrow[\text{or: acid. } K_2Cr_2O_7]{\text{acid. } KMnO_4} 2CO_2 + H_2O$$

Formation of acyl chloride

(a) carboxylic acid + PCl_3

$$3\underset{\text{propanoic acid}}{CH_3CH_2COOH} + PCl_3 \xrightarrow{\text{heat}} 3\underset{\text{propanoyl chloride}}{CH_3-CH_2-\overset{O}{\overset{\|}{C}}-Cl} + H_3PO_3$$

(b) carboxylic acid + PCl_5

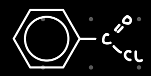
$$\underset{\text{propanoic acid}}{CH_3CH_2COOH} + PCl_5 \longrightarrow \underset{\text{propanoyl chloride}}{CH_3-CH_2-\overset{O}{\overset{\|}{C}}-Cl} + POCl_3 + HCl$$

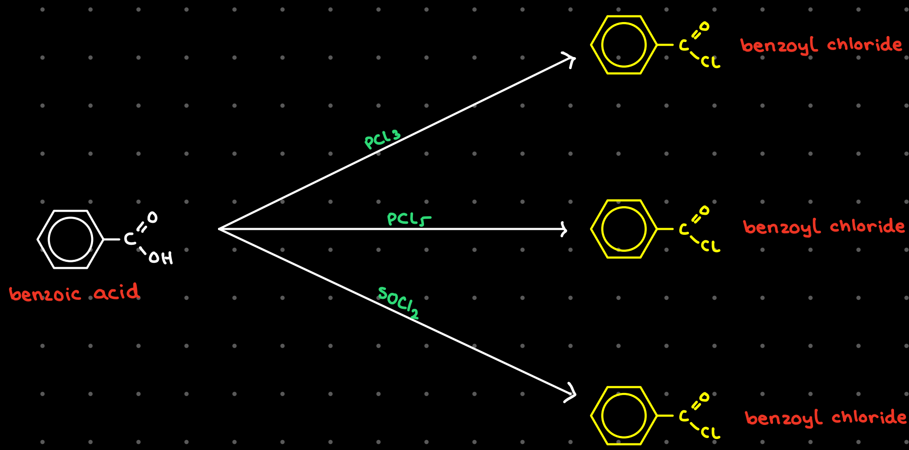
(c) carboxylic acid + $SOCl_2$

$$\underset{\text{propanoic acid}}{CH_3CH_2COOH} + SOCl_2 \longrightarrow \underset{\text{propanoyl chloride}}{CH_3-CH_2-\overset{O}{\overset{\|}{C}}-Cl} + SO_2 + HCl$$

★ only difference between the three reactions is the by-products

Formation of benzoyl chloride $\Longrightarrow$





★ only difference between the three reactions is the by-products

Reactions of acyl chlorides

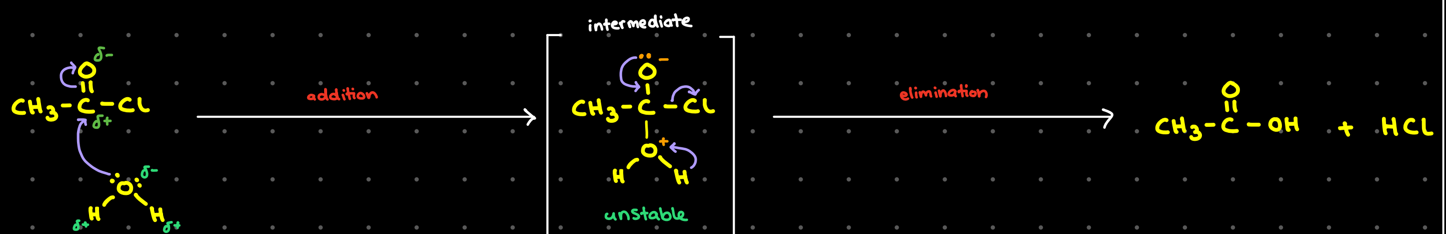
(1) hydrolysis
(2) esterification

(1) hydrolysis

$$CH_3-\overset{O}{\overset{\|}{C}}-Cl \;+\; H-OH \longrightarrow CH_3-\overset{O}{\overset{\|}{C}}-OH \;+\; HCl$$

acyl chloride → carboxylic acid

reaction mechanism : ADDITION-ELIMINATION



1) lone pair on oxygen attacks the carbonyl carbon

2) the C=O bond breaks & e⁻ pair moves to oxygen, making it negative

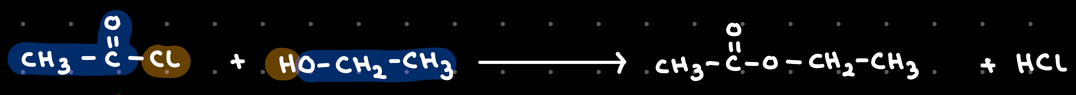
3) temporary unstable intermediate forms where carbonyl oxygen has -ve charge & water oxygen has +ve charge

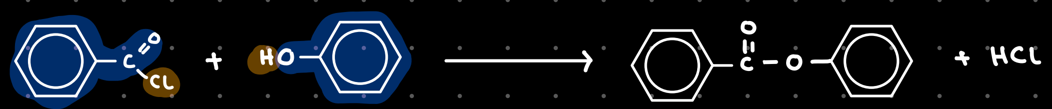
4) O⁻ lone pair drops back, kicking out the Cl⁻ (leaving group)

5) proton is removed from water oxygen

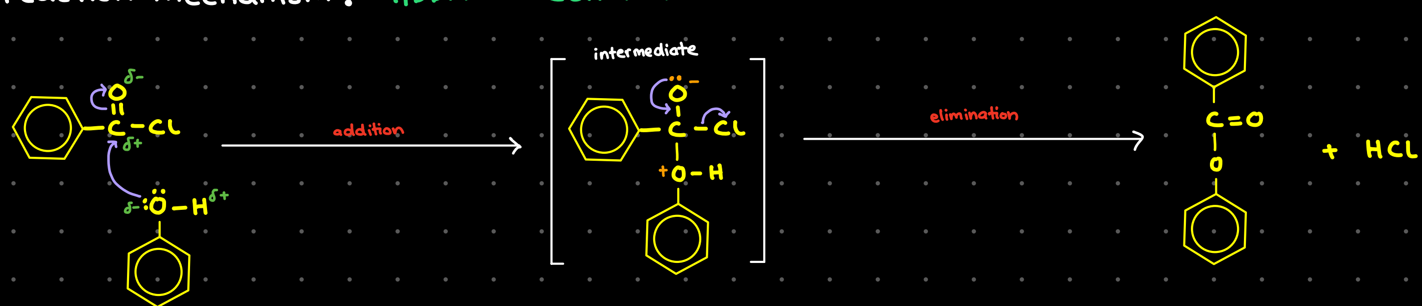
6) ethanoic acid is formed!

(2) esterification





reaction mechanism : ADDITION-ELIMINATION



1) lone pair on oxygen attacks the carbonyl carbon

2) the C=O bond breaks & e⁻ pair moves to oxygen, making it negative

3) temporary unstable intermediate forms where carbonyl oxygen has -ve charge & phenol oxygen has +ve charge

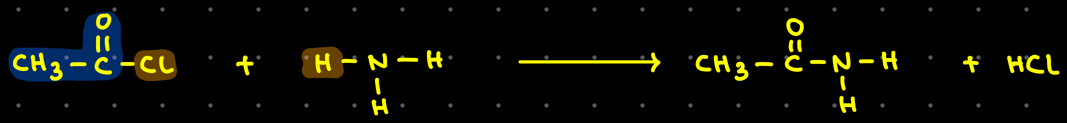
4) O⁻ lone pair drops back, kicking out the Cl⁻ (leaving group)

5) proton is removed from phenol oxygen

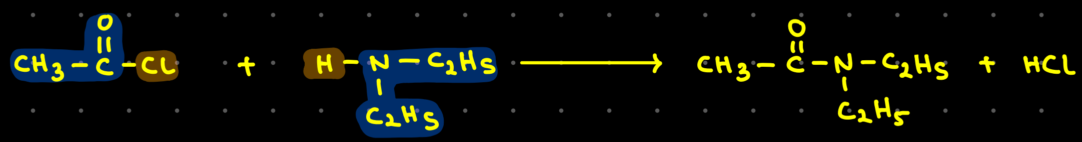
6) phenyl ester is formed !

Preparation of amides

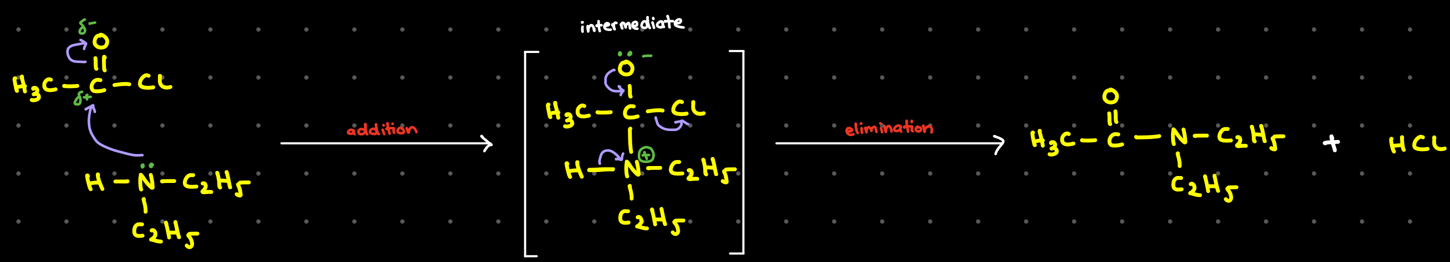
(a) acyl chloride + NH_3



(a) acyl chloride + amide



reaction mechanism : ADDITION - ELIMINATION



1) lone pair on nitrogen attacks the carbonyl carbon

2) the C=O π bond breaks & electron pair moves to oxygen, making it -ve

3) temporary unstable intermediate forms where oxygen has a -ve charge & nitrogen has a +ve charge

4) O^- lone pair drops back, kicking out the Cl^- (leaving group)

5) proton is removed from nitrogen

6) N,N-diethylamide is formed!

Reactions of amides

(1) acidic hydrolysis
(2) basic hydrolysis
(3) reduction

(1) acidic hydrolysis

$$CH_3-\overset{O}{\overset{\|}{C}}-\underset{H}{\underset{|}{N}}-H + H-OH \xrightarrow{[H^+]} CH_3-\overset{O}{\overset{\|}{C}}-OH + NH_3$$

ethaneamide — carbox. acid

NH_3 → if excess H^+ → NH^{4+} is formed !

(2) basic hydrolysis

$$CH_3-\overset{O}{\overset{\|}{C}}-\underset{H}{\underset{|}{N}}-H + NaOH \longrightarrow CH_3-\overset{O}{\overset{\|}{C}}-O^-Na^+ + NH_3$$

ethaneamide — salt

step 1 : $CH_3-\overset{O}{\overset{\|}{C}}-\underset{H}{\underset{|}{N}}-H + H-OH \longrightarrow CH_3-\overset{O}{\overset{\|}{C}}-OH + NH_3$

carbox. acid

step 2 : $CH_3-\overset{O}{\overset{\|}{C}}-OH + NaOH \longrightarrow CH_3-\overset{O}{\overset{\|}{C}}-O^-Na^+ + H_2O$

carbox. acid — salt

Nitrogen compounds

Formation of amines

(1) halogenoalkane + ammonia
(2) reduction of nitriles
(3) reduction of amides

(1) halogenoalkane + ammonia

formation of primary amine :

CH_3-CH_2-Br (halogenoalkane) + $H-\underset{|}{\overset{}{N}}-H$ with H below (ammonia (ethanolic)) $\xrightarrow[\text{pressure}]{\text{heat under}}$ $CH_3-CH_2-NH_2$ (primary amine) + HBr

formation of secondary amine :
if halogenoalkane is in excess

CH_3-CH_2-Br (halogenoalkane) + $H_2N-CH_2-CH_3$ (primary amine) $\xrightarrow[\text{pressure}]{\text{heat under}}$ $CH_3-CH_2-\underset{\underset{H}{|}}{N}-CH_2-CH_3$ (secondary amine) + HBr

formation of tertiary amine :
if halogenoalkane is in more excess

CH_3-CH_2-Br (halogenoalkane) + $CH_3-CH_2-\underset{\underset{H}{|}}{N}-CH_2-CH_3$ $\xrightarrow[\text{pressure}]{\text{heat under}}$ $CH_3-CH_2-\underset{\underset{C_2H_5}{|}}{N}-CH_2-CH_3$ (tertiary amine) + HBr

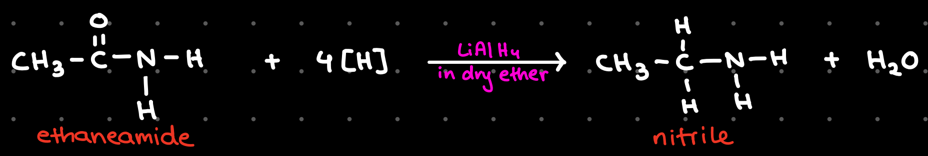
(2) reduction of nitriles

How will we prepare the nitrile ?

halogenoalkane CH_3-CH_2-CL + $KCN_{(alc)}$ $\xrightarrow[\text{reflux}]{\text{heat}}$ CH_3-CH_2-CN nitrile

$$CH_3-CH_2-C\equiv N + 4[H] \xrightarrow[\text{in dry ether}]{LiAlH_4} CH_3-CH_2-\underset{\underset{H}{|}}{\overset{\overset{H}{|}}{C}}-\underset{\underset{H}{|}}{N}-H$$

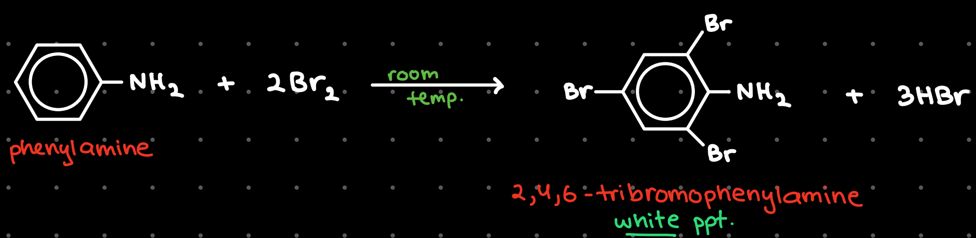
(3) reduction of amides



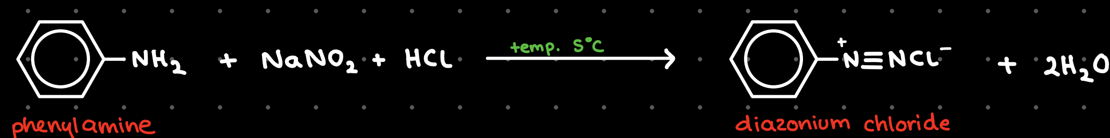
Reactions of phenylamine

(1) bromination
(2) diazotisation

(1) bromination

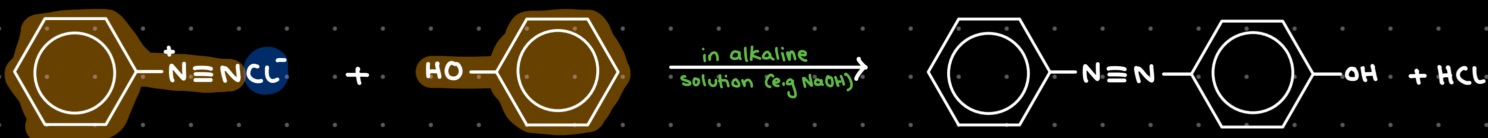


(2) diazotisation



when temp. goes >10°C :

Formation of dyes

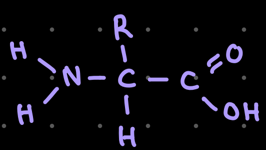


diazonium chloride

phenol

azo dye

Amino acids



→ condensation polymerisation

- ↳ monomers: amino acids
- ↳ forms polypeptides/proteins
- ↳ amide linkage
- ↳ biodegradeable polymers (can be hydrolysed)

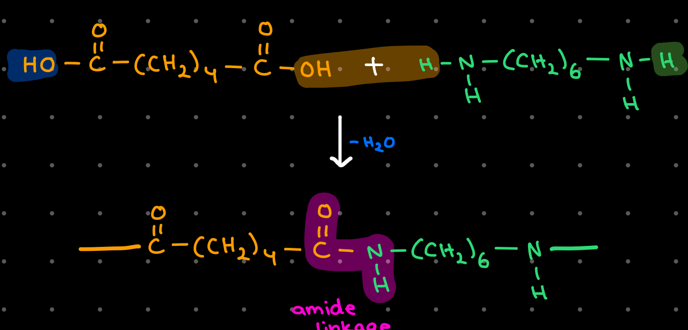
* dipeptide : made of 2 amino acids
* polypeptide : made of many amino acids

e.g. Nylon - 6,6

monomers :

1) $HO-\overset{O}{\overset{\|}{C}}-(CH_2)_4-\overset{O}{\overset{\|}{C}}-OH$ hexan-1,6-dioic acid

2) $H-\underset{H}{\underset{|}{N}}-(CH_2)_6-\underset{H}{\underset{|}{N}}-H$ hexane-1,6-diamine



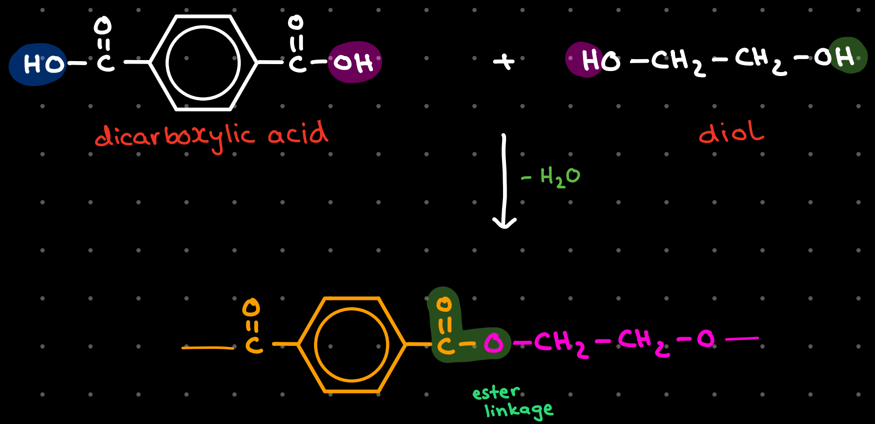
i'm bored of polymerisation so i can't be bothered to write more

Formation of polyesters

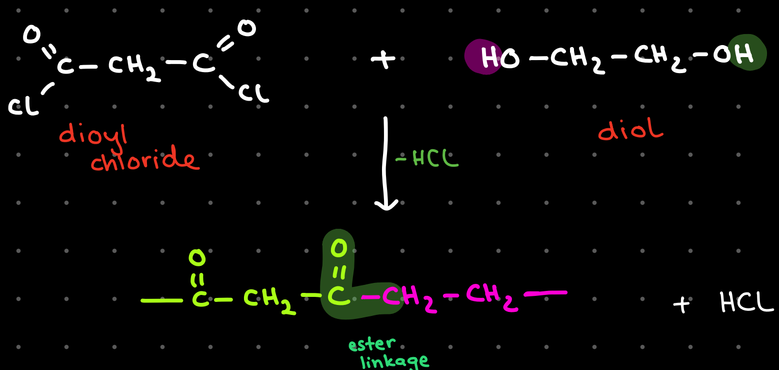
(1) diol + dicarboxylic acid
(2) diol + dioyl chloride
(3) hydroxycarboxylic acid

(1) diol + dicarboxylic acid

formation of terylene :



(2) diol + dioyl chloride



(3) hydroxycarboxylic acid